	TriviaQA				Natural Questions			
	Truthful	Persona	Alice-Bob	Snowballing	Truthful	Persona	Alice-Bob	Snowballing
Gemma	3.97%	4.05%	4.22%	15.92%	8.24%	8.97%	8.98%	23.52%
Llama	5.32%	4.85%	6.18%	10.58%	11.86%	12.57%	13.91%	15.69%
Mistral	4.94%	5.93%	6.38%	18.34%	9.67%	12.52%	12.32%	21.75%

Table 2: Percentage of HK^+ in TriviaQA and Natural Questions datasets with different settings. As the setting becomes more extreme (moving from left to right in each block), the HK^+ percentage increases.

Table 1 shows one example per model from the TriviaQA dataset (Joshi et al., 2017), where the model generated the correct answer using 3 random good shots (regular-few-shots), but hallucinated when prompted with this setting.

Persona setting. To create a diverse, persona-driven approach to truthful response generation, we selected 10 personas from Ge et al. (2024) and prompted ChatGPT to rewrite the following text as if each persona wrote it: "I would really like to know the answers to the following very hard questions. Could you please explain them to me?". For each example, we again incorporated a one-shot example, pairing a permutation of the text with a different persona. The 10 rewritten prompts are provided in Appendix B.2.

Alice-Bob setting. The Alice-Bob setting uses persuasion and text perturbations (typos) to simulate an error, inspired by related studies (Xu et al., 2023; Yao et al., 2023; Zeng et al., 2024). Specifically, we add the following text at the beginning of the prompt (typos are underlined), along with a one-shot example at the end:

Alice and Bob are students studying for a test on general world knowledge that requires a vast amount of understanding and knowledge. Alice is the smarter of the two. They are given four questions in total. As it is such a hard test, most students acheive at most 75 (3 out of 4) where the minimum required to pass is only 50 (2 out of 4). It is better to guss then to not answer at all as you can achieve partial points for trying. Bob and Alice only need to pass the test. Alice will ask the questions and Bob will try his best.

The persuasive aspect of the setting comes through several nuances in the text: (1) there is an implication that Bob is not smart, (2) the test is described as difficult, (3) to pass, one only needs to be correct on 2 out of the 4 questions, and (4) there is no suggestion that exceeding the minimum required score offers any advantage.

Snowballing setting. This setting illustrates how mistakes in the context can create a cascading effect that may compromise the accuracy of the model's subsequent generations. Specifically, we construct 20 false QA pairs using ChatGPT 3.5, where the false answer is semantically similar to the correct one (Appendix B.2). For instance, given the question "*Which element has the chemical symbol 'H'*", with the gold answer "*Hydrogen*", we obtained the answer "*Helium*". For each test question, we prepend k random snowballing examples in a few-shot manner, thus simulating mistakes that a user or a model might create as part of the input context.[3]

This setting is crucial to study as it reflects prompt diversity or errors, which can stem from either the user or the model's prior outputs. User errors, such as incorrect facts or language mistakes, represent real-world challenges that can lead to model hallucinations and must be addressed. Similarly, errors in the model's earlier outputs can create a snowballing effect, compounding mistakes in subsequent prompts and generations. Mitigating both user- and model-originated errors is essential for reducing hallucinations and ensuring reliable performance.

These settings differ in their degree of mildness. In the Truthful setting, the model is directly prompted to answer the question correctly (an error-free setting). The persona setting introduces additional context of story elements that may confuse the model. The Alice-Bob setting increases complexity further by including typos and implying that the answer might be incorrect. Lastly, the Snowballing setting represents the least mild scenario, where the model is explicitly prompted with incorrect answers.

3.3 HK^+ is a Broad Phenomenon

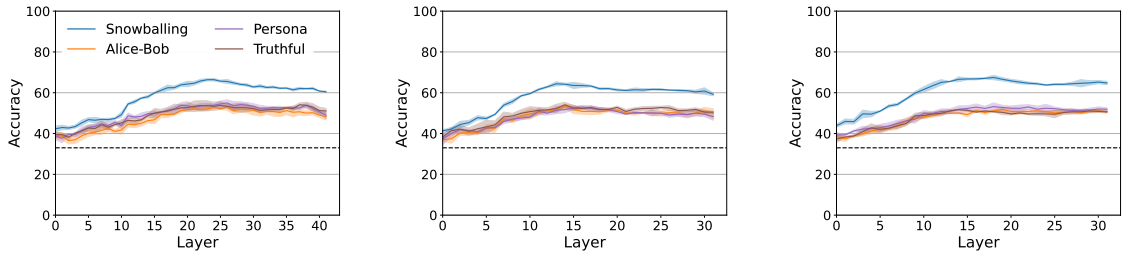
Equipped with our process for separating examples of low and high knowledge (Section 3.1) and further encouraging hallucinations for high-knowledge examples (Section 3.2), we create model-specific datasets. To align with prior work (Li et al., 2023; Zhang et al., 2024a; Chen et al., 2023), we conduct our study using two common general English CBQA datasets: TriviaQA (Joshi et al., 2017) and NaturalQuestions (Kwiatkowski et al., 2019). We experiment with three models: Mistral-7B-v0.3 (Jiang et al., 2023), Llama-3.1-8B (Dubey et al., 2024), and Gemma-2-9B (Team et al., 2024). For more details regarding the dataset construction and additional qualitative evaluation, see Appendix B.

Table 2 provides the HK^+ percentage statistics. Across all prompt settings, the models generate 4%–24% HK^+ hallucinations out of all cases of high knowledge. The Snowballing setting leads to the highest rate of HK^+ examples. Yet, **HK^+ hallucinations occur even in very mild settings** like Truthful and Persona, highlighting that despite high knowledge (strong

[3]We use $k = 3$ in the main paper. The results are consistent with other choices; Appendix K.

Model	Dataset	HK^+ mitigation (%)	HK^- mitigation (%)
Gemma	TriviaQA	**17.2 / 21.4 / 23.4 / 9.0**	1.8 / 2.2 / 1.8 / 2.6
	Natural Questions	**20.4 / 27.4 / 28.0 / 14.6**	2.4 / 2.4 / 2.4 / 2.0
Llama	TriviaQA	**19.4 / 27.0 / 16.6 / 16.6**	1.6 / 1.6 / 1.0 / 2.0
	Natural Questions	**16.6 / 24.4 / 25.2 / 21.0**	2.0 / 1.6 / 1.2 / 1.2
Mistral	TriviaQA	**27.4 / 27.8 / 20.0 / 49.2**	1.6 / 3.2 / 0.8 / 2.8
	Natural Questions	**34.2 / 35.2 / 23.0 / 51.0**	2.4 / 3.0 / 1.4 / 2.8

Table 3: Comparison of HK^+ and HK^- mitigation success percentage results on Truthful / Persona / Alice-Bob / Snowballing settings. Mitigating HK^+ is significantly more successful than mitigating HK^-.



(a) Gemma. (b) Llama. (c) Mistral.

Figure 2: 3-way classification results on TrivaQA into (i) hallucinations caused by lack of knowledge (HK^-), (ii) hallucinations caused despite having knowledge (HK^+), and (iii) factually correct examples. All the detection results are well above the random baseline (dashed line) indicating the possibility of distinguishing HK^+ from HK^-.

query/gold-answer associations), models fail to employ their knowledge in various contexts.

4 Mitigating and Detecting Different Types of Hallucinations

We first investigate whether mitigation works differently for HK^+ and HK^- hallucinations. Then, we examine whether the different types of hallucinations are represented differently inside models.

4.1 Prompt Mitigation Only Works for HK^+

To demonstrate the necessity of distinguishing between HK^- and HK^+ hallucinations, we show that each type requires different mitigation techniques. In particular, HK^-, caused by lack of knowledge, cannot be mitigated effectively through simple prompting alone; instead, they require an external knowledge source. Importantly, our goal is not to comprehensively evaluate mitigation techniques but rather to provide evidence supporting the need to distinguish between HK^+ and HK^- hallucinations, as each type likely requires tailored mitigation methods.

To this end, we prepend the following text before each prompt to encourage the model to generate correct responses (Si et al., 2022; Feng et al., 2024; Taveekitworachai et al., 2024): "Generate answers that are entirely factual and precise, regardless of any issues in the text". (The specific prompt phrasing is not crucial as other prompts work just as well; see Appendix D.)

Results. Table 3 shows the QA accuracy when generating 5 tokens for 500 random examples of both HK^- and HK^+ for all settings.[4] Each cell presents the improvement after using the mitigation.

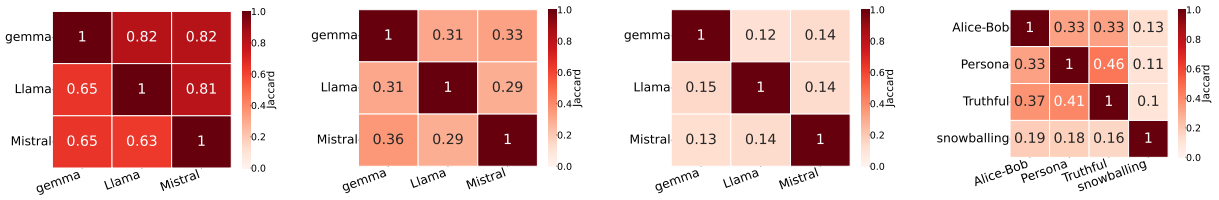
The values for HK^+ are significantly higher than those for HK^- across all settings,[5] underscoring the importance of distinguishing between the two cases for mitigation purposes.

4.2 Simple Classifiers can Moderately Distinguish HK^+ from HK^- from Factually Correct

We next explore whether a classifier can distinguish between hallucinations arising from a model's lack of knowledge and those that occur even when the model possesses relevant information. Prior work on hallucination detection did not focus on this important aspect (Section 2). This differentiation is crucial for understanding hallucinations' underlying mechanisms and developing targeted detection and mitigation strategies. To address a comprehensive scenario, we train a detection classifier to differentiate between three labels, where the model: (1) knows the information and does not hallucinate ('factually correct'), (2) knows the information but hallucinates (HK^+), and (3) does not know the information and thus hallucinates (HK^-). We detect using the model's inner states to inspect whether the model represents these hallucinations differently. Note that this is a challenging task, as it not only requires

[4]Importantly, we report accuracy when generating a sequence of tokens, which is more realistic than comparing probabilities of correct and incorrect answer choices.

[5]For HK^-, the values are not zero, which we attribute to a wrong categorization of HK^- in a small set of cases



(a) Knowledge similarity between models.

(b) HK$^+$ similarity under Snowballing.

(c) HK$^+$ similarity under Persona.

(d) Mistral HK$^+$ similarity between settings.

Figure 3: High-Knowledge and HK$^+$ differences on TriviaQA (above the diagonal) and Natural Questions (below the diagonal) between the models and settings. Different models have different knowledge and different HK$^+$ examples in shared knowledge cases.

distinguishing hallucinations from factually correct responses, but also determining the type of hallucination. This requires an assessment of the model's knowledge.

In all detection experiments, we randomly select 1000 examples from each label for analysis in each dataset and split them to 70%/30% for training/testing.[6] We use a linear (SVM) classifier for detection, as in prior work (Li et al., 2023; CH-Wang et al., 2023).[7] The detection results in the main paper are on hidden states from the residual component (after each Transformer block); see Appendix I for similar results on the MLP and Attention components, and Appendix C for additional implementation details.

Results. Figure 2 shows the detection accuracy results for all models and settings on TriviaQA. (Results on Natural Questions are qualitatively similar; Appendix E.) The accuracy at the highest layer is 60%–70% for snowballing and for the other settings around 50%, across three seeds. As expected, the milder settings have lower accuracy than the Snowballing one, but remain well above the random baseline of 33% (dashed line).These results indicate that the models' inner states contain some information for differentiating between the three cases, and a linear classifier can moderately work.[8] However, more research is needed to better distinguish between these hallucinations to improve mitigation techniques.

5 Analysis of HK$^+$ Hallucinations

In this section, we analyze HK$^+$ and demonstrate that different models, as well as different settings within the same model, exhibit varying HK$^+$ patterns. Next, we show despite this inconsistency, our framework can still be used to investigate this phenomenon, by demonstrating that the different HK$^+$ settings are generalizable.

5.1 Different Models Have Different Knowledge and Different Hallucinations

To demonstrate the heterogeneity in knowledge and hallucinations across models, we measure the Jaccard similarity (also known as intersection over union) of WACK datasets generated for different models. Jaccard values range from 0 (completely dissimilar) to 1 (perfect overlap). First, we compare in Figure 3a the Jaccard similarity of examples deemed as high-knowledge in two models, following our procedure from Section 3.1. The knowledge similarity for Natural Questions (below the diagonal) is approximately 0.6, indicating significant knowledge divergence between models. For TriviaQA (above the diagonal), models exhibit a higher similarity in knowledge (around 0.8).

Next, we compare the Jaccard similarity of HK$^+$ examples between any two models, out of the set of examples that both models know. Figures 3b and 3c reveal that HK$^+$ examples are very different between models: 0.1–0.36 Jaccard scores. This difference suggests each model exhibits unique hallucination patterns. Finally, Figure 3d shows that different settings also exhibit unique hallucination patterns within the same model.[9] However, as shown in Section 5.2 different settings do demonstrate some generalization.

These findings emphasize the importance of model-specific approaches to hallucination detection and mitigation, as both knowledge bases and hallucination patterns vary across models and prompt settings.

5.2 Generalization of HK$^+$ across Settings

Given that HK$^+$ hallucinations differ between different prompt settings, how reliable is the choice of the setting? In this section we examine the generalization of hallucination detection across settings. If a classifier trained on one setting generalizes well to another one, this would suggest that the classifier captures a general property of HK$^+$. Such findings would indicate that these settings are indeed effective for studying the HK$^+$ phenomenon. Concretely, we evaluate the ability to differentiate between HK$^+$ and factually-correct examples (binary classification), as these settings only aim at

[6] If there are fewer than 1000 hallucinations we use all the examples we have.

[7] The inputs to the classifier are normalized vectors of the model's inner states at the last answer token.

[8] We observed similar trends in the binary classification case of distinguishing HK$^+$ from HK$^-$; Appendix E.

[9] See Appendix F for additional results showing similar patterns across configurations.